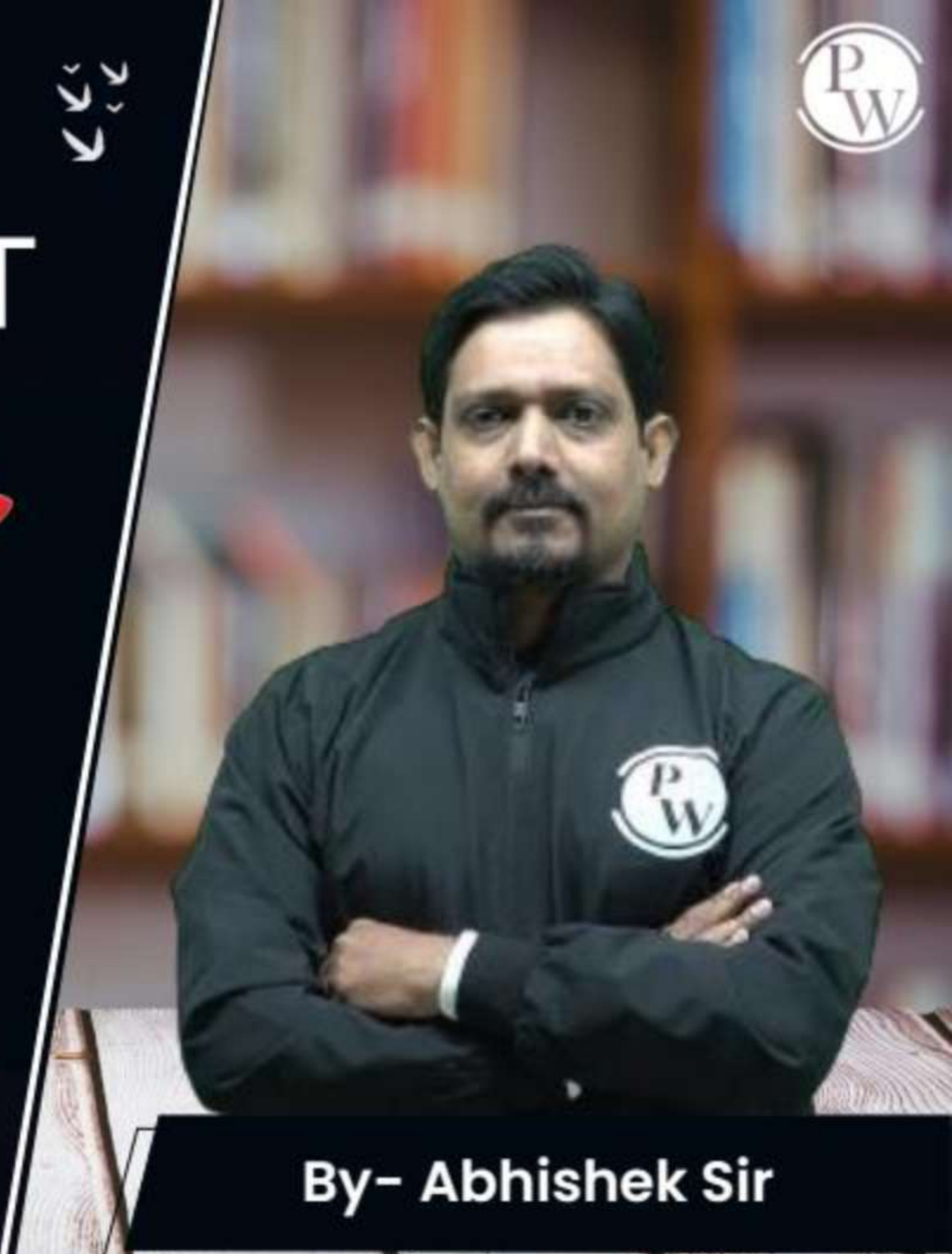
Computer Science & IT



C Programming

Function & Storage Class

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Storage class

auto

static -

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Static problem

Recursion

Types of Recursion
- C. Language
- Data structure Application
- Mathematics DM
- Analysis (Algo.)

Slide



Storage Class

* Register
* Extern



Recursion

Recursion is problem Solving Technique in which Solution of problem expresses Solution of Smaller instance of Same problem.

In programming Recursion is a function that call itself. (A function call to itself)

Slide



Recursion

Recursive ⟶ Base condition :- Termination

progress towards the base condition

How to make it smaller / larger

So that base condition is

achieved.

Slide



Recursion

Factorial of No. $\lfloor 5 = 5 \times 4 \times 3 \times 2 \times 1$

$$n! = \begin{cases} n * (n-1)! & n > 1 \\ 1 & n = 0 \,||\, n = 1 \end{cases}$$

$\lfloor 5$

$5 * \lfloor 4$

$* 4 * \lfloor 3$

$3 * \lfloor 2$

$2 * \lfloor 1$

↑ Base Condition

Slide



Recursion

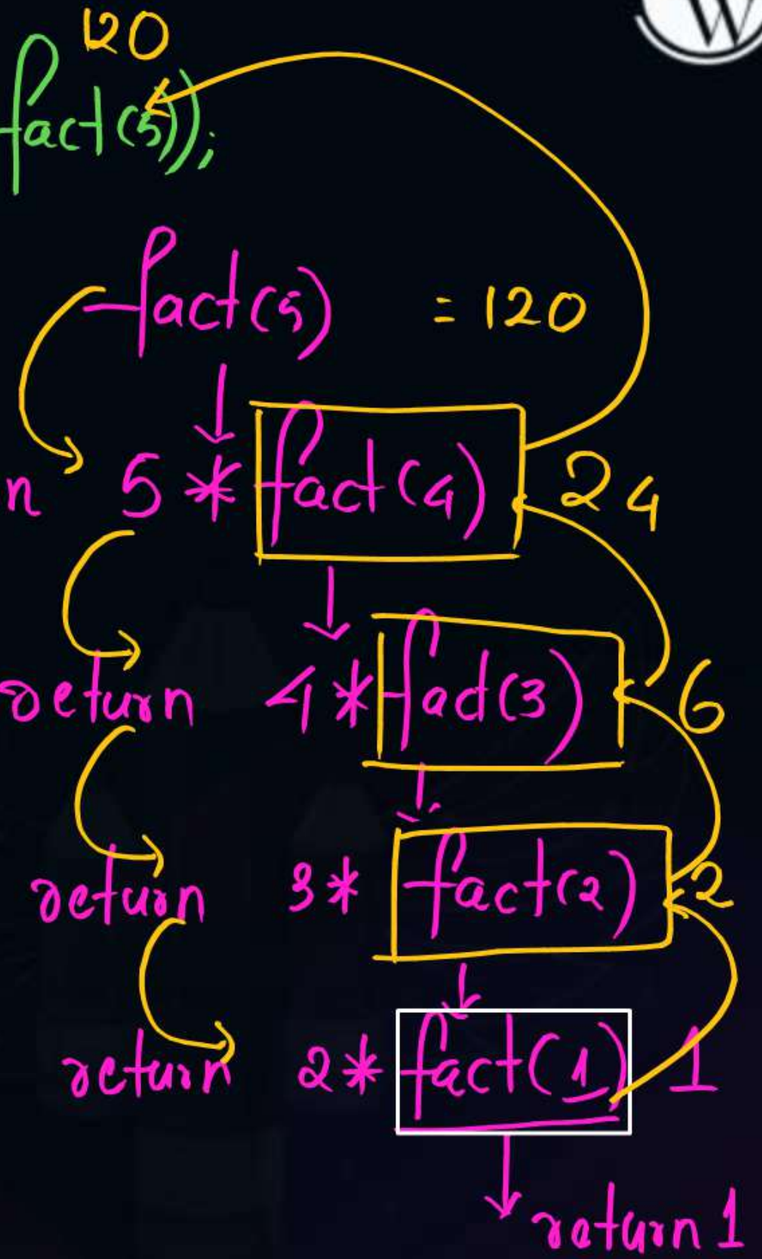
```
int fact(int n){
    if(n==0||n==1)
        return 1;
    else
        return n*fact(n-1);
}
```

fact is a function which is calling fact

Slide

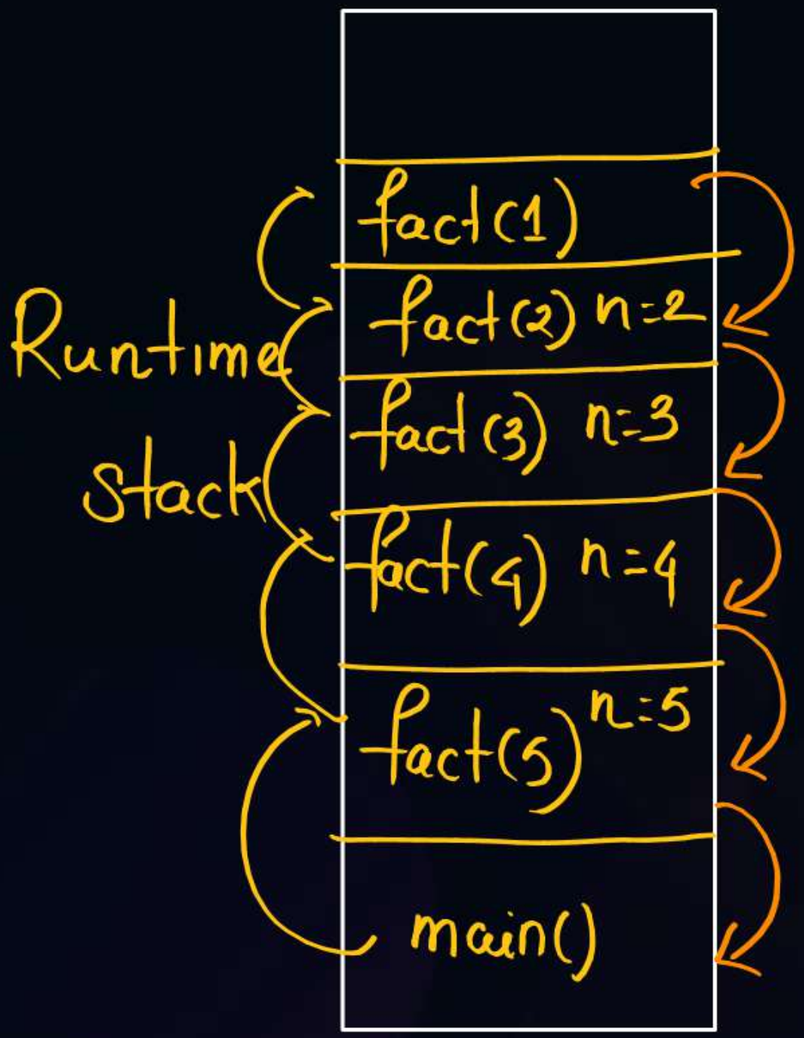
Recursion

```
#include<stdio.h>
int fact(int n){
    if(n==0||n==1)
        return 1;
    else
        return n*fact(n-1);
}
int main(){
    printf("%d",fact(5));
    return 0;
}
```



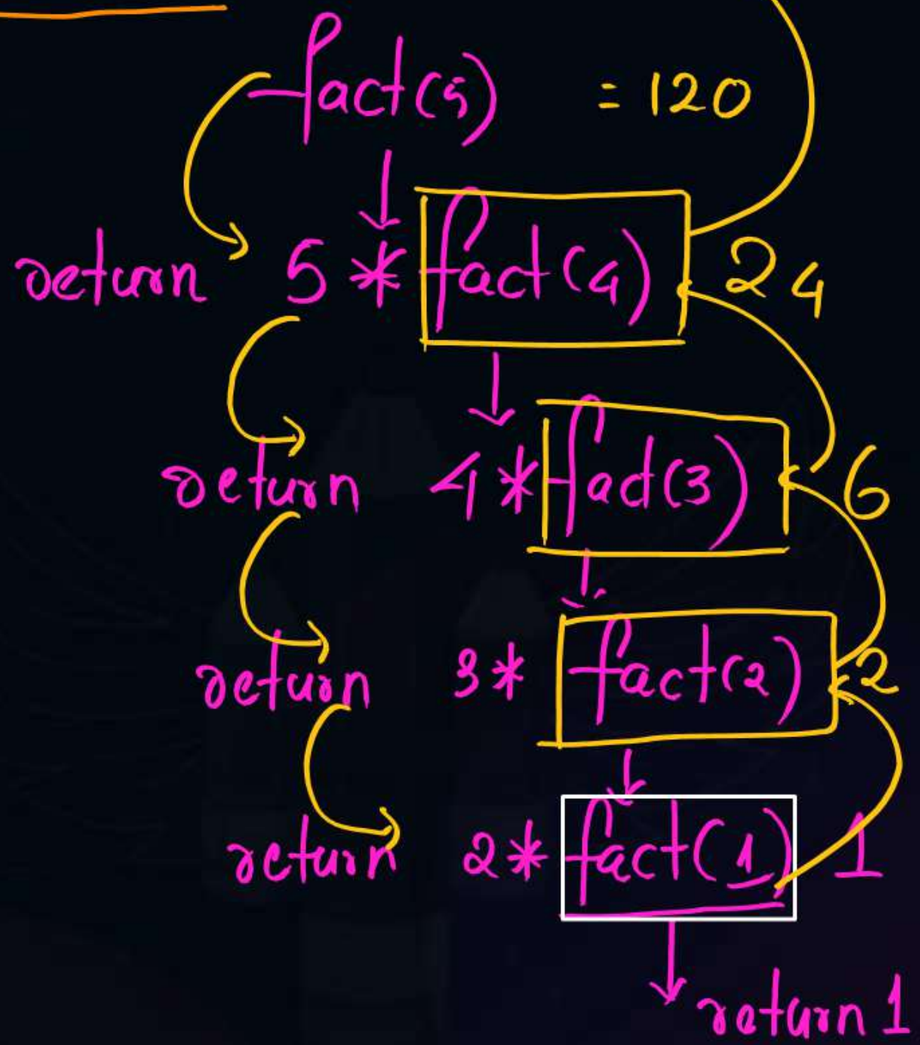
Slide

Recursion




```
int main(){
    printf("%d", fact(5));
}
```

120



Slide



Recursion

Single Recursion → Tail Recursion ∴ If Recursive call is Last statement of Recursive function then we call it as Tail Recursion.

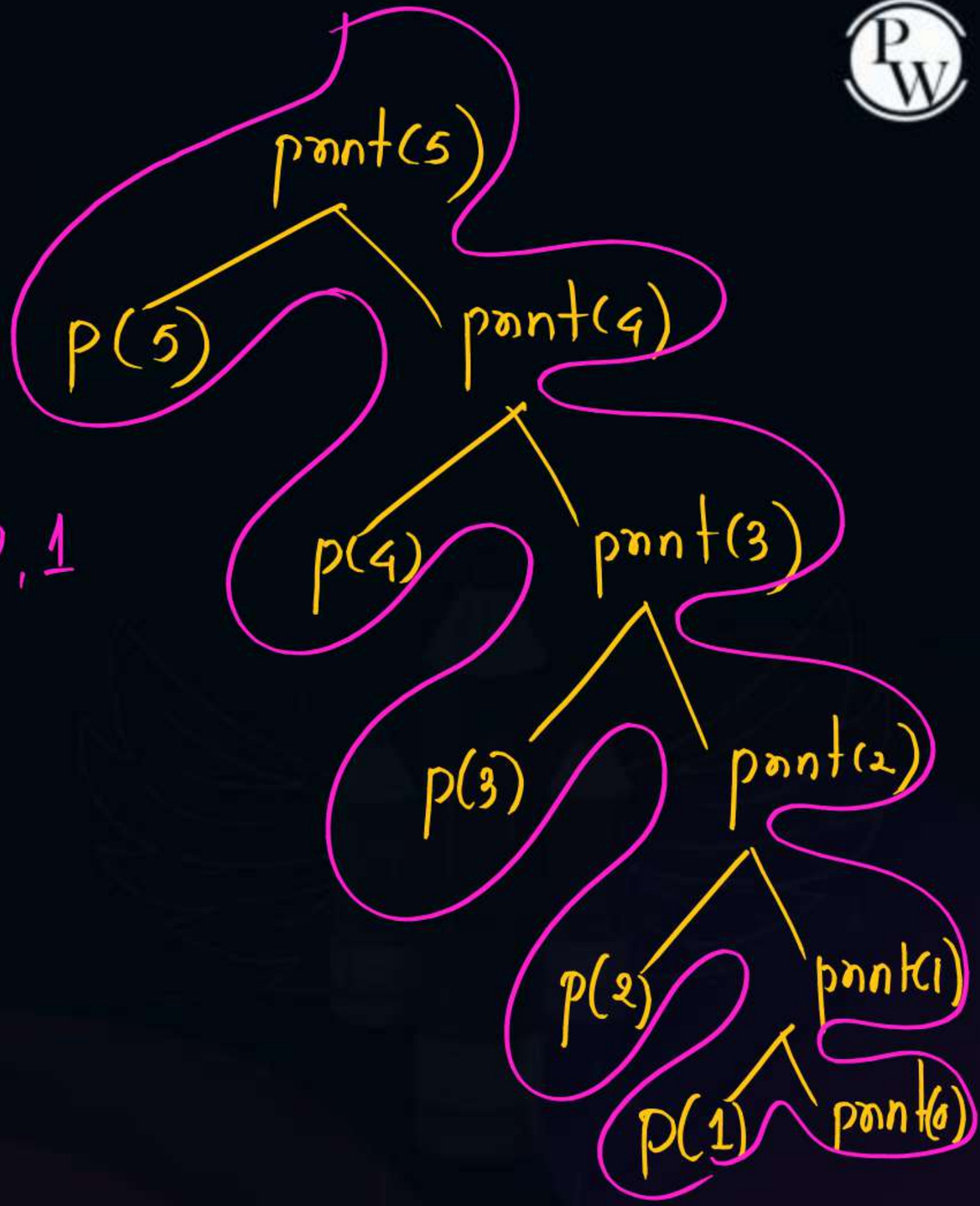


Recursion

```
#in lcude<stdio.h>
void print(int n){
      if (n <= 0)   return;
      printf("%d", n);
      print(n-1);
}
int main(){
      print(5);
      return 0;
}
```

↑ Last Statement

5, 4, 3, 2, 1



Slide



Recursion

```
#include<stdio.h>
void print(int n)    {
     if (n <= 0)   return;
     print(n-1);
     printf("%d", n);
     }

int main(){
     print(5);
     return 0;
}
```

Head Recursion

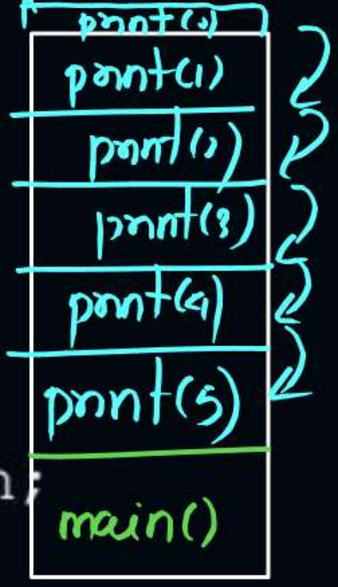
If Recursive call is first statement (after base condition) in Recursive function then we call that Recursion as Head Recursion



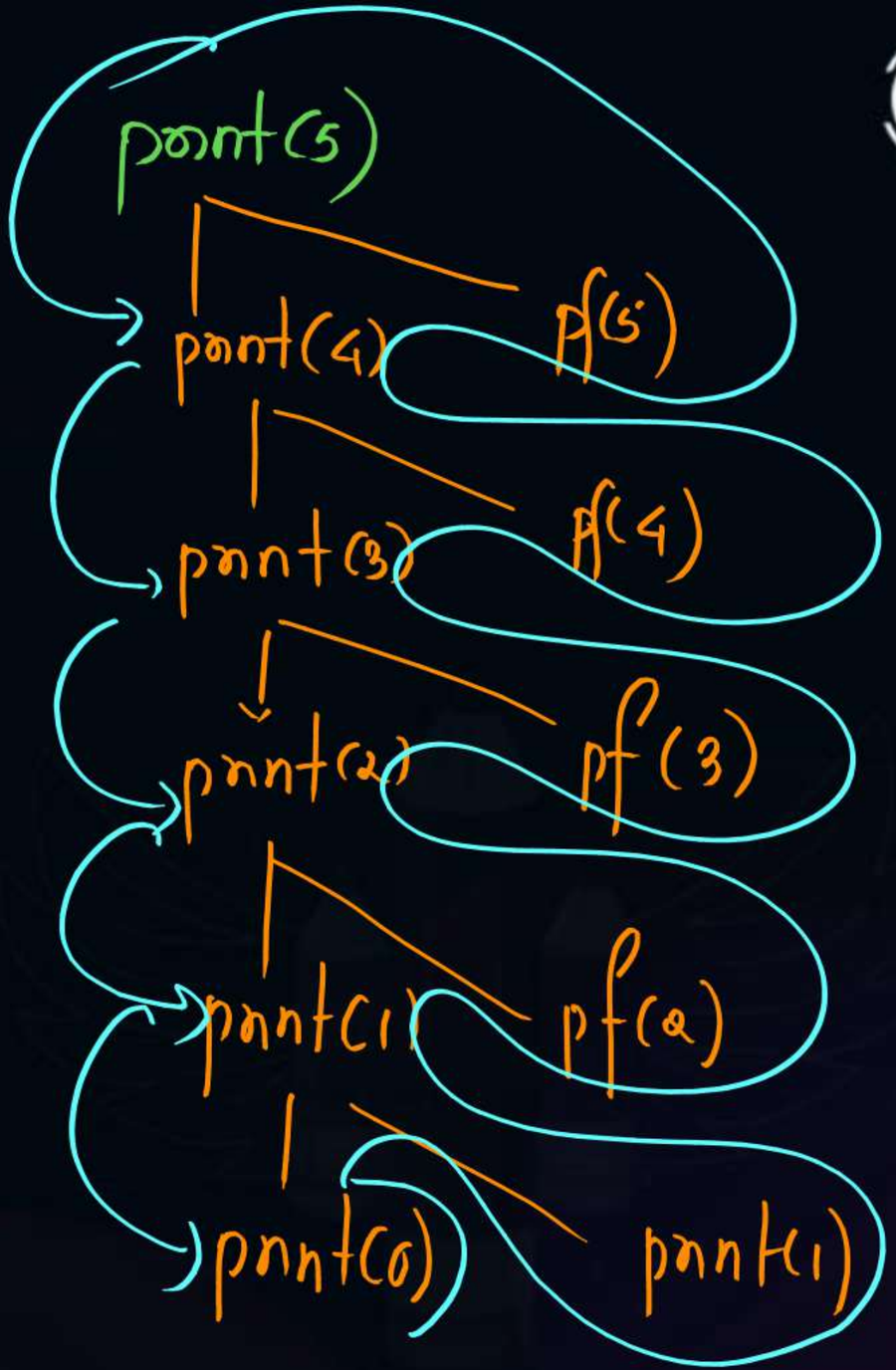
Recursion

```
#include<stdio.h>
void print(int n)    {
      if (n <= 0)   return;
      print(n-1);
      printf("%d", n);
      }

int main(){
      print(5);
      return 0;
}
```



1, 2, 3, 4, 5



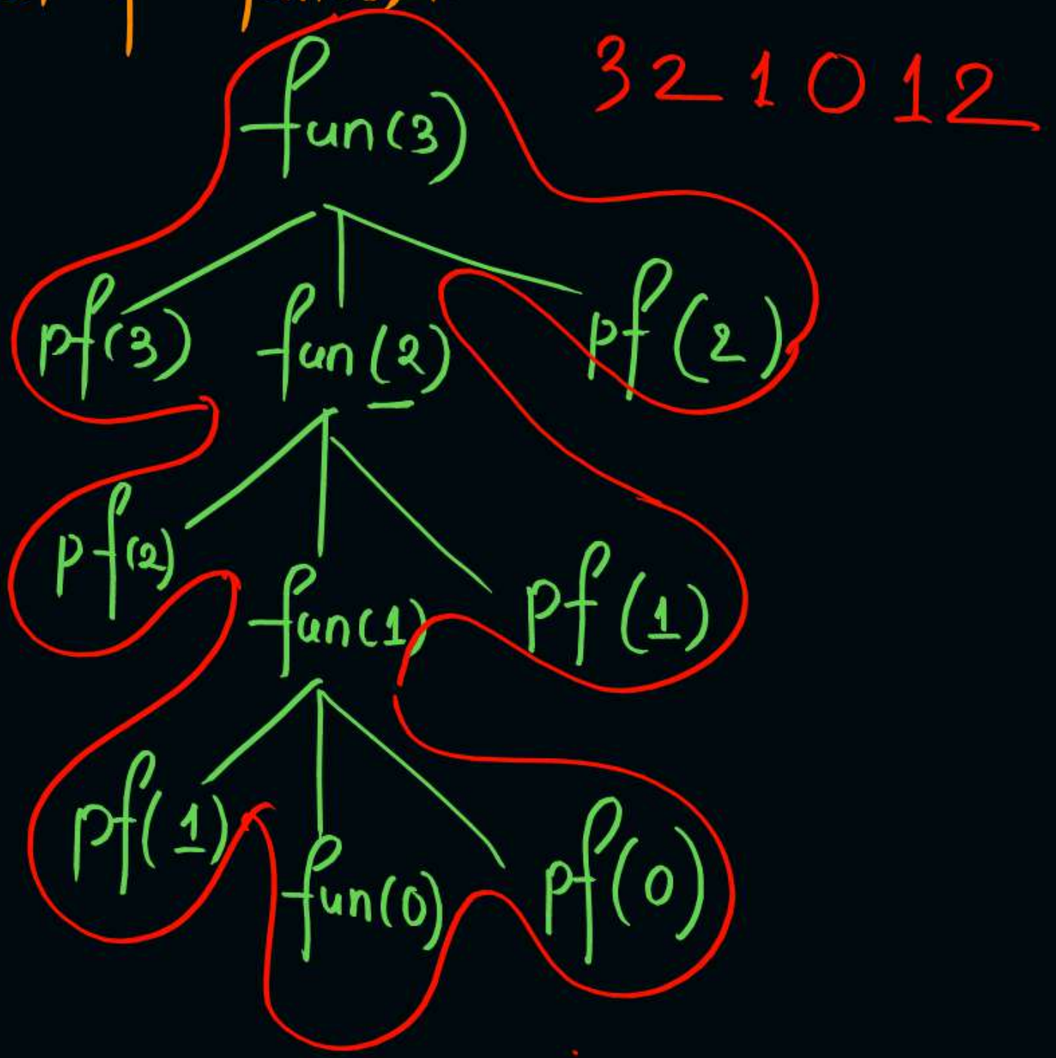
Slide

```
#include<stdio.h>
void fun(int n){
    if(n<=0) return;
    printf("%d", n);
    fun(n-1);
    printf("%d", n-1);
}
```

Neither Head Nor Tail

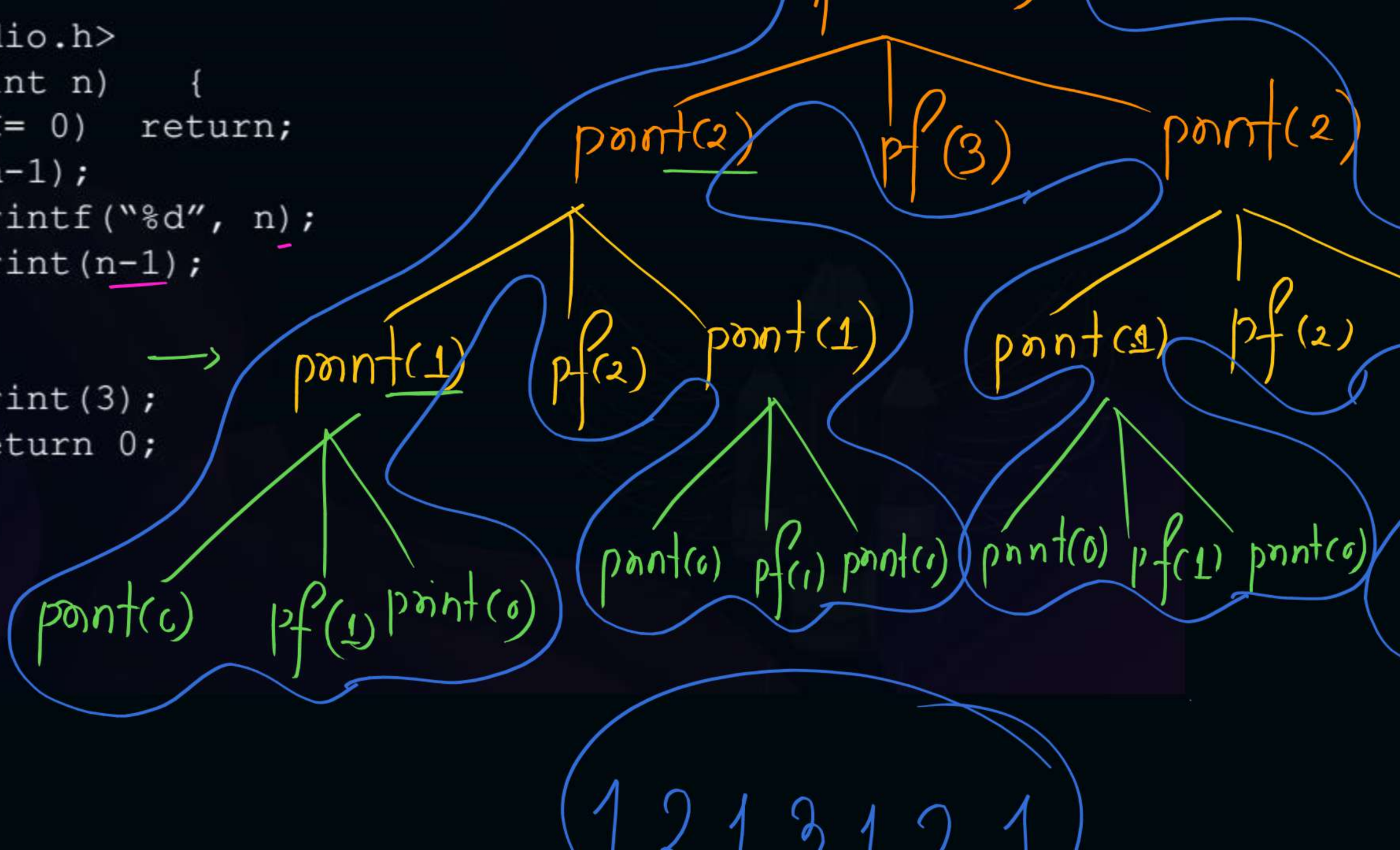
output if fun(3) is called?

3 2 1 0 1 2



```
e<stdio.h>
int(int n)    {
  (n <= 0)   return;
int(n-1);
    printf("%d", n);
    print(n-1);

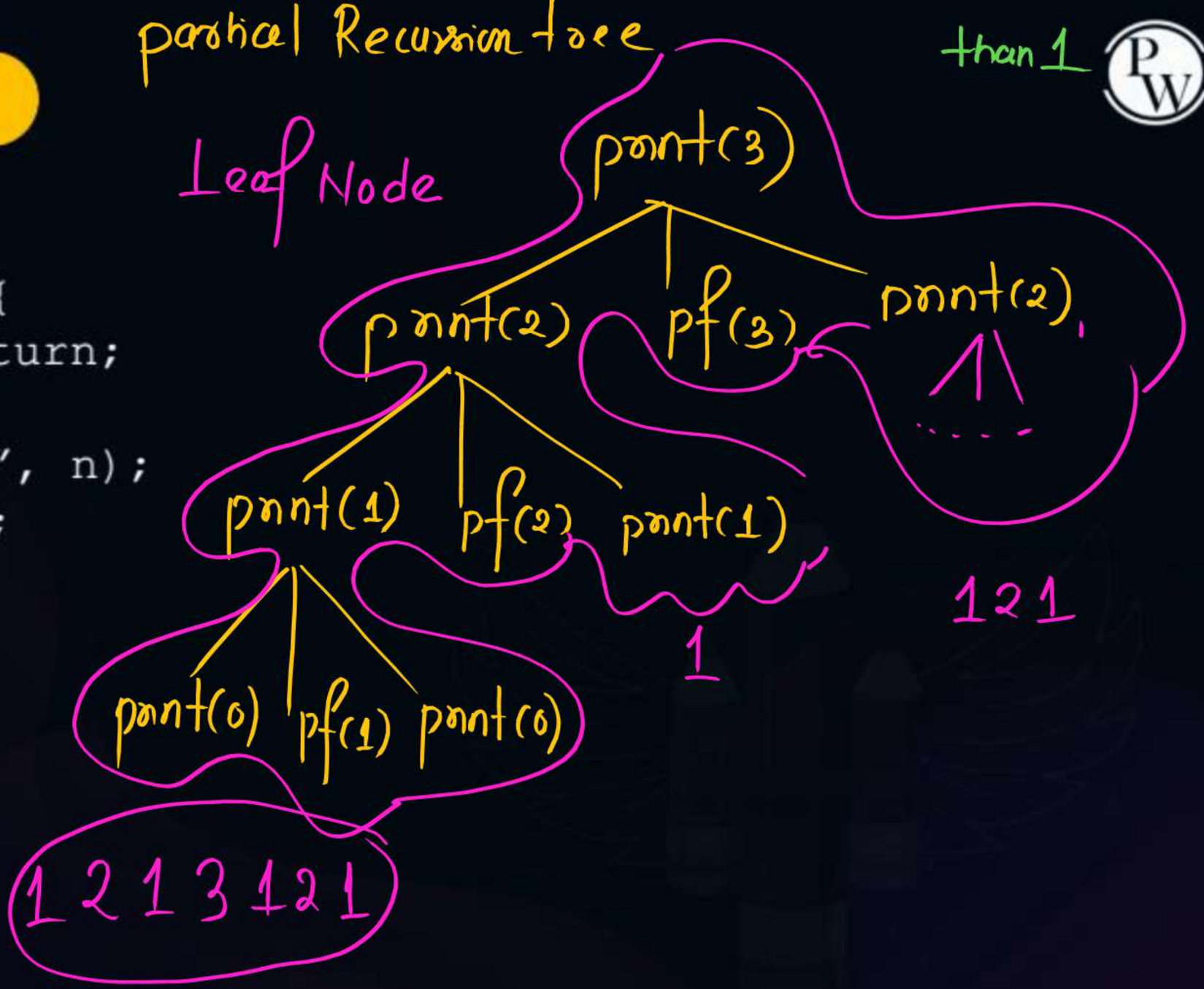
n(){
    print(3);
    return 0;
```





Recursion

```
#include<stdio.h>
void print(int n)    {
      if (n <= 0)   return;
      print(n-1);
            printf("%d", n);
            print(n-1);
      }
int main(){
            print(3);
            return 0;
}
```



Slide



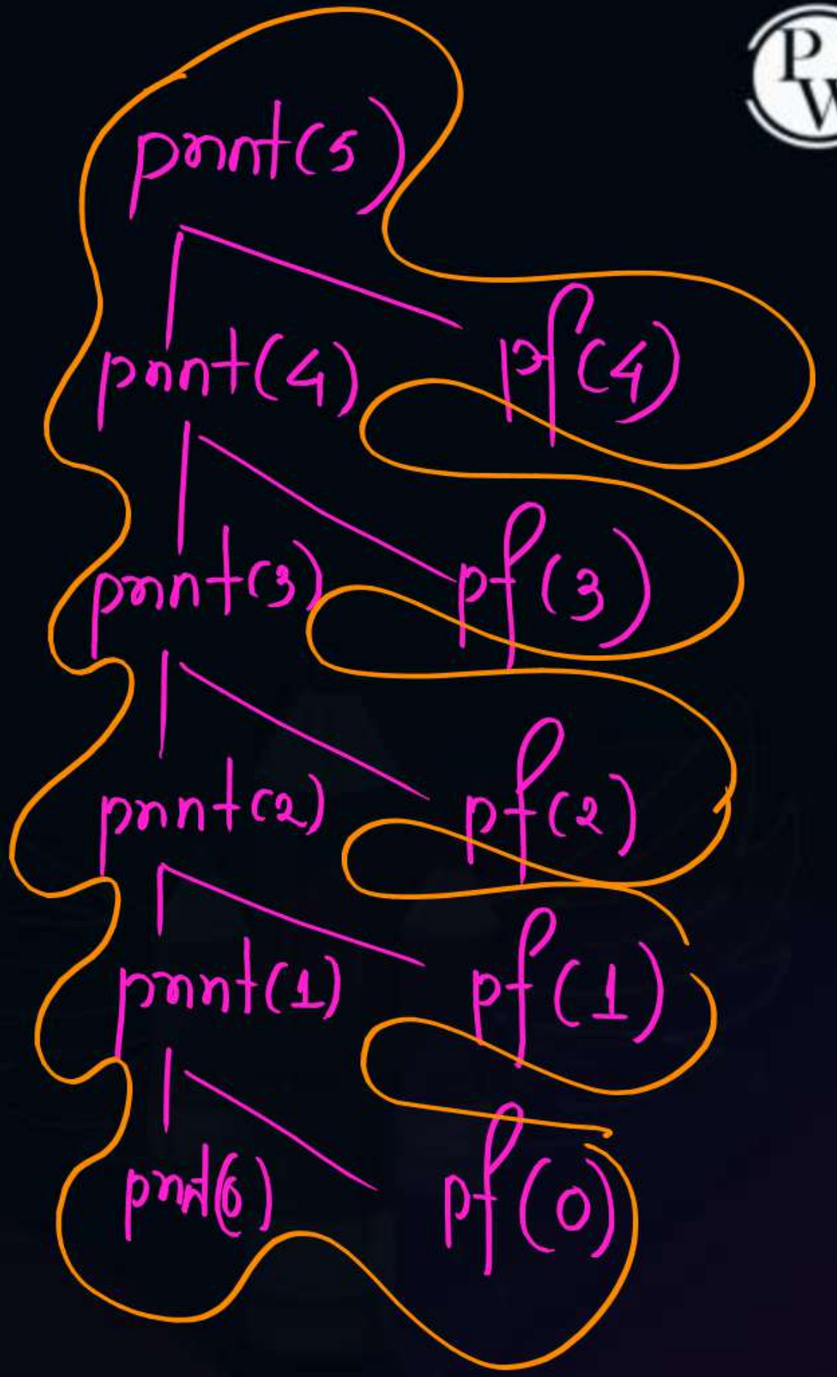
Recursion

```
#include<stdio.h>
void print(int n)    {
      if (n <= 0)   return;
      print(--n);
      printf("%d", n);
      }

int main(){
      print(5);
      return 0;
}
```

--n;
↑ modify n value

0, 1, 2, 3, 4



Slide

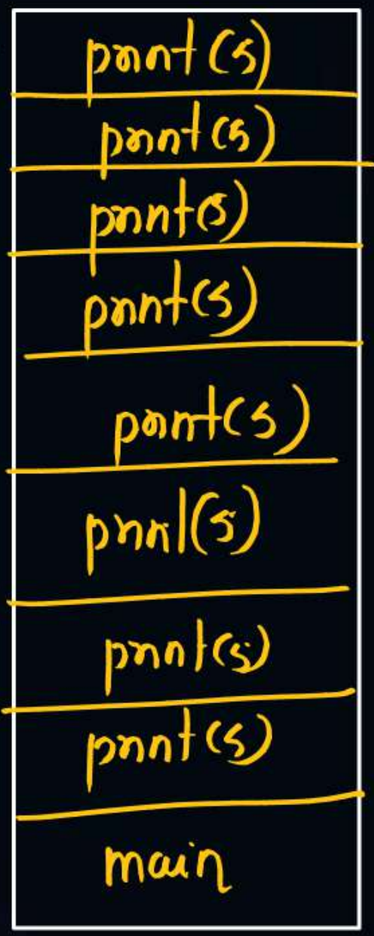


Recursion

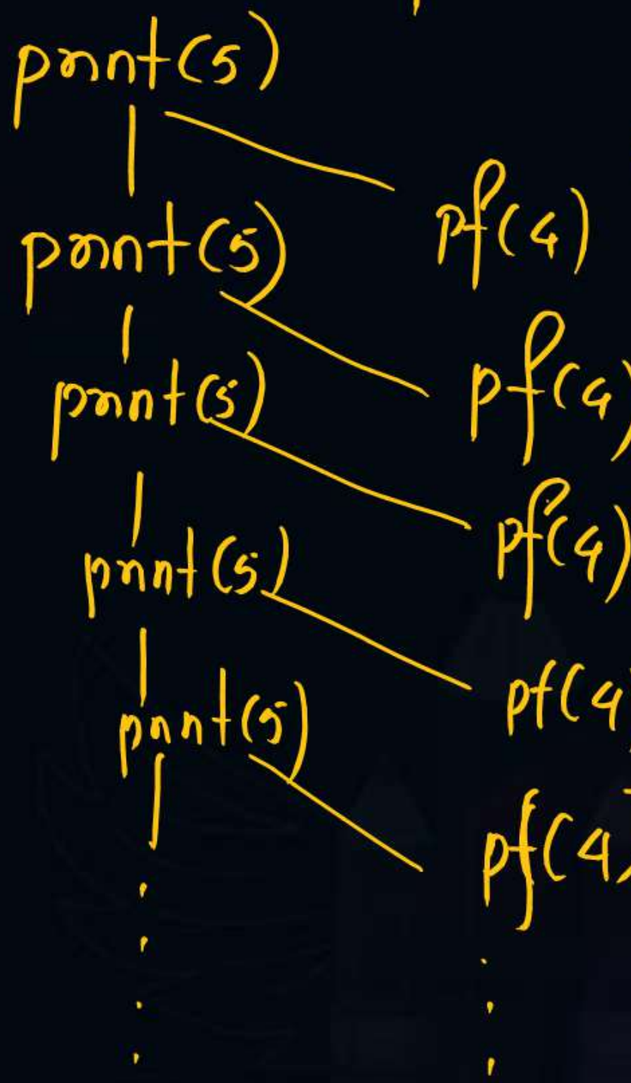
```
#include<stdio.h>
void print(int n)    {
      if (n <= 0)   return;
      print(n--);
      printf("%d", n);
      }

int main(){
      print(5);
      return 0;
}
```

Stack ki galti bhar gayi



Stack overflow



* abnormal Termination

* Segmentation fault.

Slide



Recursion

#Q. Consider the following program

```
#include<stdio.h>
int foo(int n){

      if (n<=9)
          return n;
    else
          return n%10+foo(n/10);
}

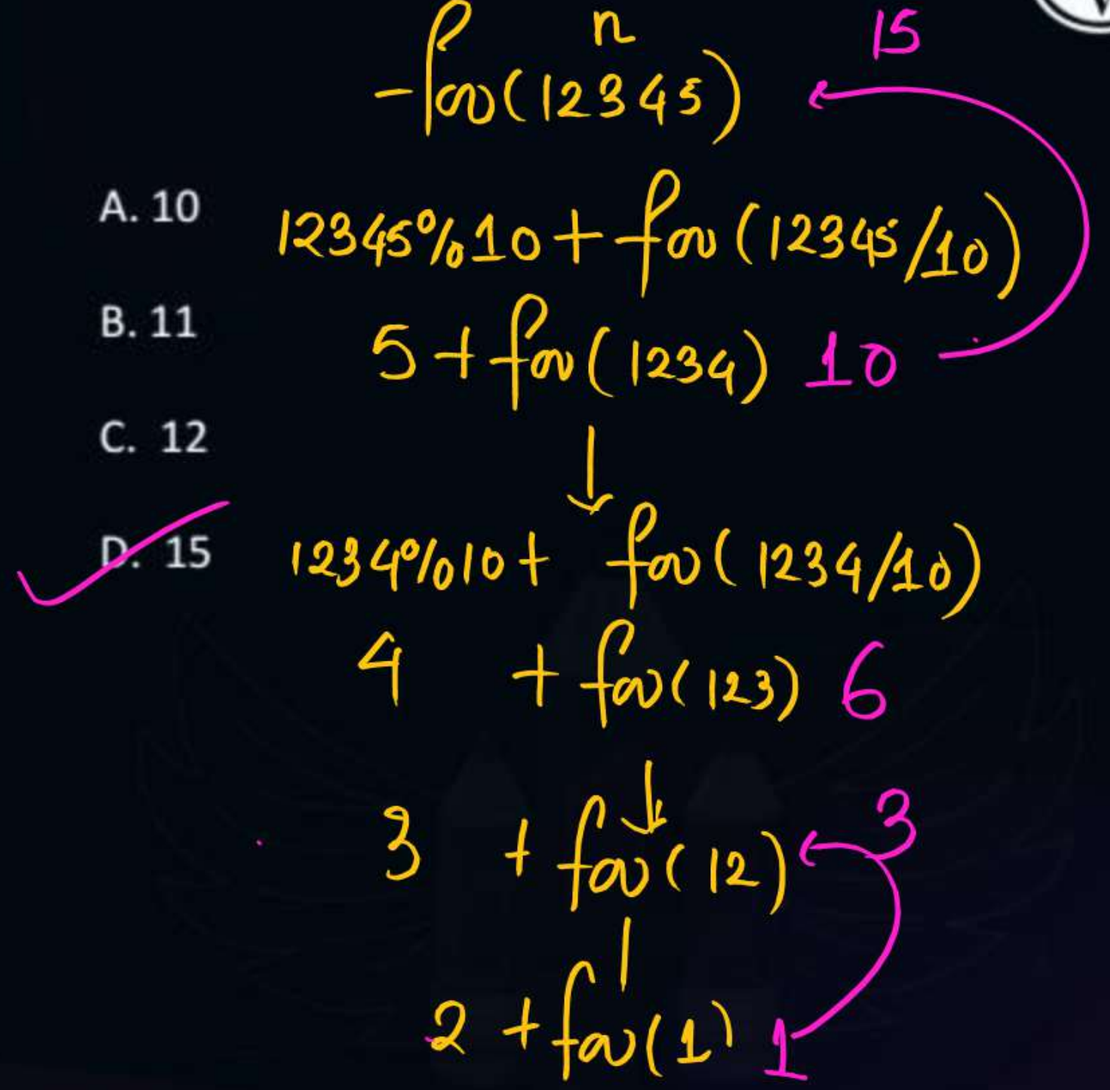
int main(){
      printf("%d", foo(12345));
      return 0;
}
```

A. 10

B. 11

C. 12

D. 15



Slide



Recursion

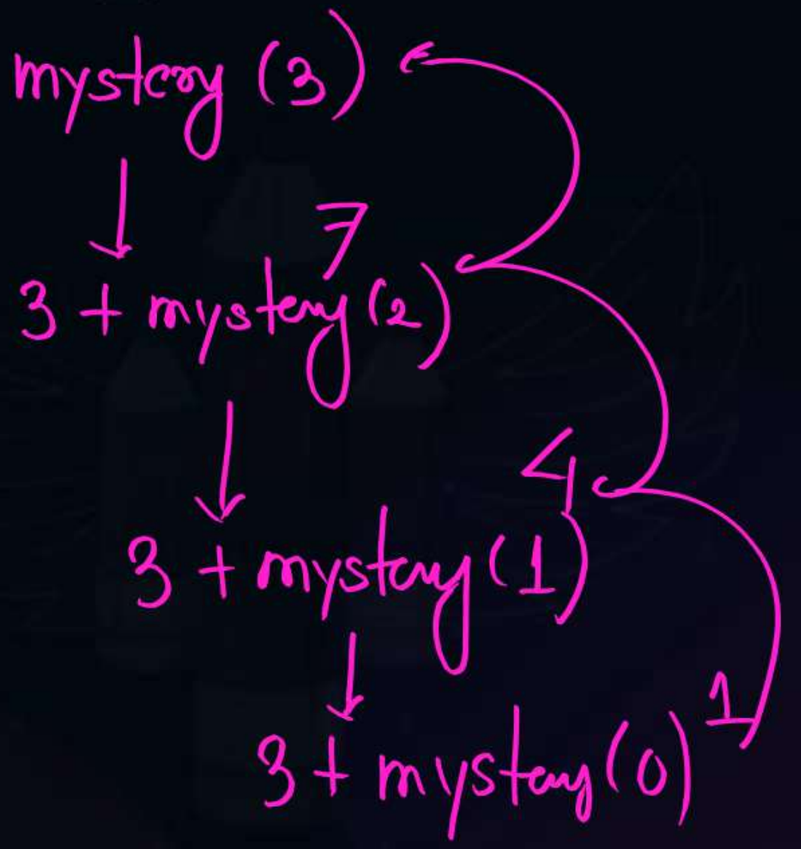
#Q. Consider the following program

```
int mystery(int n) {
        if (n <= 0)
                return 1;
        else
                return 3 + mystery(n - 1);
}
```

What is the output the program when mystery(3) is called

(A) 9
(B) 10
(C) 11
(D) 12

10



Slide



Recursion

#Q. Consider the following program

```
void Dosomething(int n) {
      if (n > 1)
            Dosomething(n-1);
      for (int i = 0; i < n; i++)
            printf("*");
      printf("\n");
}
```

The number of stars will print if the Dosomething(5) is called ?

HW

NAT



Recursion

#Q. Consider the following program

```
void quiz(int i)
    {
       if (i > 1)
       {
          quiz(i / 2);
          quiz(i / 2);
       }
       printf("*");
    }
```

What value will be returned by the call quiz(5)

(A) 8
(B) 7
(C) 3
(D) 2



2 mins Summary

Topic — Recursion

Topic — Recursion tree

Topic — partial Recursion

Topic

Topic

Slide

THANK - YOU

